

MCA SEMESTER-I
School of Computer and Systems Sciences,
Jawaharlal Nehru University, New Delhi -110067
Computer Fundamental (CS-402)

Total Marks: 20

Date: 07.02.2022

Time: 2hr

All Compulsory

5. Express the following function in a) some of minterms and b) product of maxterms.

d) $F(x,y,z)=1$

e) $F(w,x,y,z)=y'z+wx'y+wxz'+w'x'z$

f) $F(x,y,z)=(xy+z)(y+xz)$

3 5

6. For the function ; $F=xy+xy'+y'z$

e) Obtain the truth table

f) Implement the function using 2-input gates

g) Implement the function using 2-input gates after minimizing.

h) Compare the number of gate required in b) and c)

5

7. Implement 3-bit Odd function.

5

8. Simplify the following function F using don't care condition d in :

$$F=w'(x'y+x'y'+xyz)+x'z'(y+w)$$

$$D=w'x(y'z+yz)+wyz$$

• Implement the function using 2-input gates

$$\bar{w}\bar{x}+yz+\bar{x}y$$

5

$\frac{w}{x}$
 $\bar{A}+\bar{B}$

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Computer Fundamental (CS-402) Mid-I
Date: 08.09.2024

Total Marks: 20

Time: 2hr

All Compulsory

1. Convert the Hexadecimal 2AC5 to decimal, octal and Binary bases. 2
2. Simplify the Boolean function to a minimum number of literals: $x'y' + xy + x'y$ 2
3. Find the complement of : $(x+y'+z)(x'+z')(x+y)$ 2
4. Implement half subtractor. 2
5. Simplify the function: $F = \Sigma(1, 4, 5, 6, 12, 14, 15)$. Implement the function without simplification and after simplifying to check the number of gates in each case. 4
6. Implement BCD to excess three code converter. 4
7. Implement Full adder using decoder. 4

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Date: 07.10.2025

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Time: 2hr

All Compulsory

- i. a) By means of a timing diagram similar to Fig. 1(a), show the signals of the outputs F1 in Fig.1(b) as functions of the inputs x, y and z.

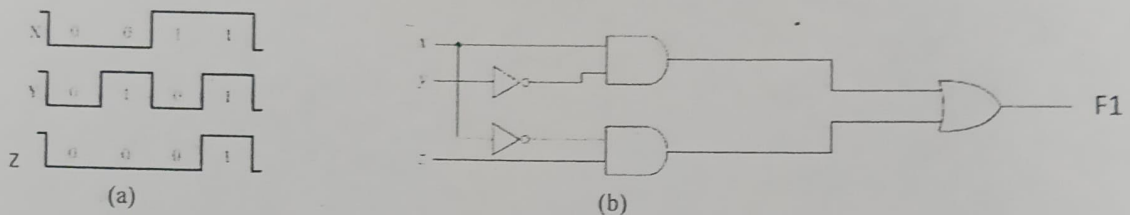
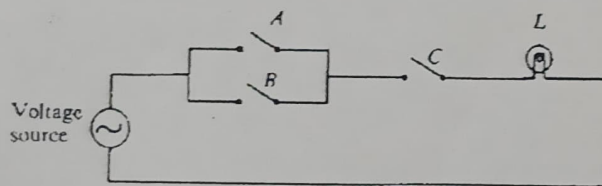


Figure 1: a) Input waveform signals b) Logic circuit

- b) Express the switching circuit in binary logic notation.



2.5+2.5=5

2. Design a combinational circuit for BCD-Excess-3 coder converter. 5
3. Design BCD-to-decimal decoder. 5
4. Design 4 to 2 Priority encoder. 5